

Variables and conditionals

Name: _____ Date: _____

Work on paper without running the code. Show your reasoning.

1. Assignment and memory

```
int a = 4;
int b = a;
a = a + 3;
b = b - 1;
```

Draw the final memory boxes for a and b. Explain why changing a does or does not change b automatically.

2. Values across frames

```
int total = 0;

void draw() {
    int count = 0;
    total = total + 1;
    count = count + 1;
    println(total + ", " + count);
}
```

What is printed by the first two calls to draw()? Explain the difference between the two variables.

3. Scope

```
void setup() {
    int score = 10;
}

void draw() {
    println(score);
}
```

Will this compile? Explain why, and describe one correction that lets draw() print the value initialized in setup().

Variables and conditionals

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4. AND and boundary values

```
mouseX > 100 && mouseX <= 200
```

Give true or false for each mouseX value. Explain the boundary cases.

100: _____ 150: _____ 200: _____ 201: _____

5. Separate decisions or one chain?

Each version starts with a fresh `int x = 2`. What final value does each version print? Explain why the results differ.

<pre>// Version A int x = 2; if (x > 0) { x = 0; } if (x == 0) { x = 5; } println(x);</pre>	<pre>// Version B int x = 2; if (x > 0) { x = 0; } else if (x == 0) { x = 5; } println(x);</pre>
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6. Write a condition

A player may enter only when room is 0 and they have either a key or a pass. `hasKey` and `hasPass` are Boolean variables.

Write the condition inside `if (...)`. Then give one set of values that allows entry and one that prevents it.
